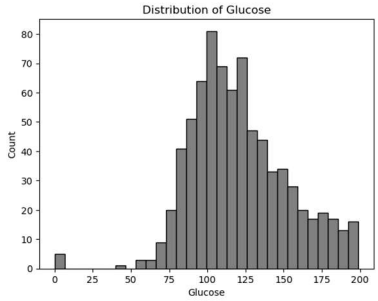
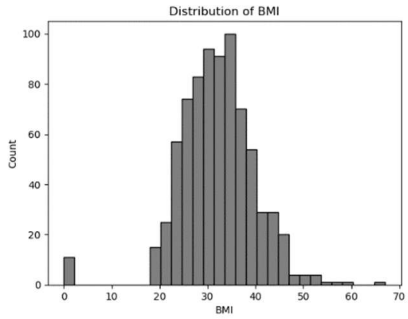
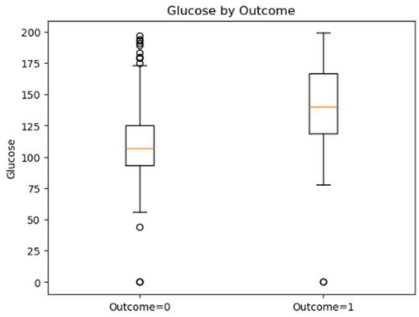
1. Data Integrity & Quality Scan





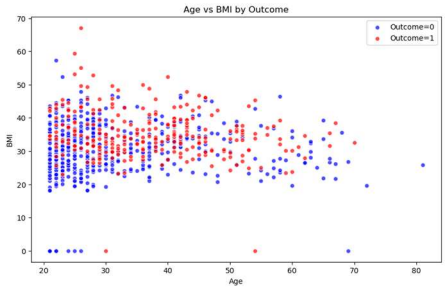
- **Findings:** The Missing Value Matrix shows no explicit NaNs. However, Histograms reveal **hidden missing values** (zeros) in Glucose, BMI, and Blood Pressure. These anomalies must be imputed before Cycle 2.

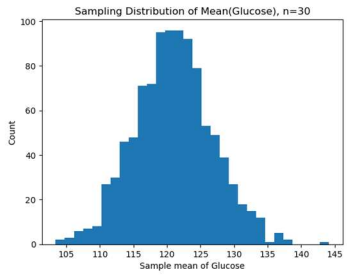
2. Distribution & Outlier Analysis



- **Findings: Glucose** and **BMI** are near-normal, while **Age** is right-skewed. Boxplots indicate multiple upper-level outliers in Insulin and DPF, and confirmed "0" outliers at the bottom of several scales.

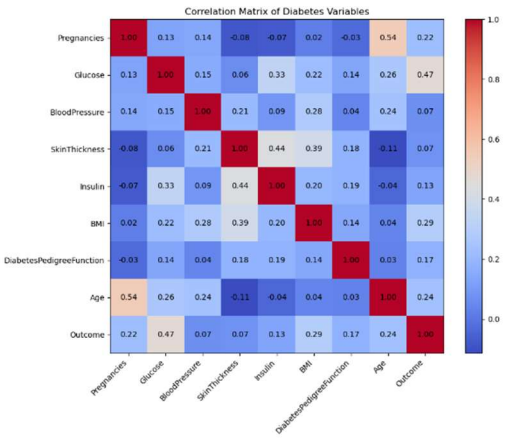
3. Bivariate Insights (Key Predictors)





- **Glucose Impact:** A clear distributional shift is observed; diabetic patients have significantly higher median glucose levels.
- **Age vs. BMI:** Scatter plots show that high BMI is a strong risk indicator across all age groups, though orange clusters (Outcome=1) are denser in the 30+ age range.

4. Correlation & Multicollinearity



- **Primary Correlate: Glucose** has the highest positive correlation with the Outcome.
- **Secondary Correlate:** BMI and Age.
- **Note:** High correlation between Age/Pregnancies suggests potential redundancy.

5. Conclusions & Next Steps

- **Conclusion: Glucose** is the most reliable predictor. **Obesity (BMI)** and **Age**